أكاديمية طويق
Tuwaiq Academy

UNIT 2 · ASSIGNMENT 3

Eigenvalues & Eigenvectors

TEAM ASSIGNMENT REPORT

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Based on the accompanying solved Jupyter Notebook

Executive Summary

This report documents six missions on eigenvalues and eigenvectors, combining manual derivations with numerical verification using NumPy. For the matrix A = [[2, 1], [1, 2]], the eigenvalues are 3 and 1 with eigenvectors proportional to $[1, 1]^T$ and $[1, -1]^T$. We interpret the corresponding geometric transformations, verify diagonalization, and demonstrate the role of eigenvalues in Principal Component Analysis (PCA).

Aim and Method

The assignment requires: (1) manual calculations, (2) verification of numerical results using NumPy, and (3) an answer to the AI Insight question for each mission. NumPy and Matplotlib are used to calculate results and illustrate the geometric meaning of the transformations. Matrix vectors are represented as columns.

$$A = [\,2 \quad 1\,;\,1 \quad 2\,]$$

Mission 1 — Discovering Eigenvectors

Given $v_1 = [1, 1]^T$ and $v_2 = [1, -1]^T$, we compute each matrix-vector product manually:

$$Av_1 = [\,2(1)+1(1)\,;\,1(1)+2(1)\,] = [3, 3]^T = 3v_1$$

$$Av_2 = [\,2(1)+1(-1)\,;\,1(1)+2(-1)\,] = [1, -1]^T = 1v_2$$

Both vectors remain on their original lines: v_1 is stretched by a factor of 3, while v_2 is unchanged. Thus both satisfy the eigenvector equation $Av = \lambda v$. NumPy verifies A @ v_1 = [3, 3] and A @ v_2 = [1, −1].

AI Insight | Most vectors change direction because the transformation mixes coordinates. Eigenvectors are special nonzero vectors that lie along preserved lines. For this matrix, those lines are spanned by $[1, 1]^T$ and $[1, -1]^T$.

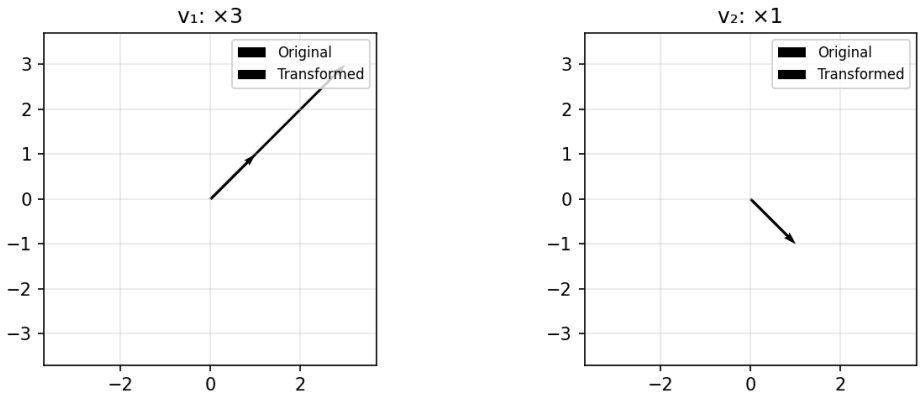


Figure 1. Both eigenvectors retain their line after transformation.

Mission 2 — Finding Eigenvalues

Eigenvalues are found by solving the characteristic equation $\det(A - \lambda I) = 0$.

$$A - \lambda I = [\ 2-\lambda \quad 1\ ;\ 1 \quad 2-\lambda\ ]$$

$$\det(A - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3$$

$$(\lambda - 3)(\lambda - 1) = 0 \ \Rightarrow\ \lambda_1 = 3, \lambda_2 = 1$$

NumPy np.linalg.eig(A) returns eigenvalues 3 and 1 (order can vary). Substituting either eigenvalue into det(A − λI) gives 0, confirming the manual result.

AI Insight | An eigenvalue measures how the associated eigenvector is scaled. A positive eigenvalue maintains orientation; a negative eigenvalue reverses it. Its absolute value is the scaling factor.

Mission 3 — Finding Eigenvectors

For λ = 3, solve (A − 3I)v = 0:

$$[\ -1 \quad 1\ ;\ 1 \quad -1\ ]\ [x, y]^T = 0 \ \Rightarrow\ -x + y = 0 \ \Rightarrow\ y = x$$

$$\text{Choose } x = 1\text{: } v_1 = [1, 1]^T$$

For λ = 1, solve (A − I)v = 0:

$$[\ 1 \quad 1\ ;\ 1 \quad 1\ ]\ [x, y]^T = 0 \ \Rightarrow\ x + y = 0 \ \Rightarrow\ y = -x$$

$$\text{Choose } x = 1\text{: } v_2 = [1, -1]^T$$

Normalize by dividing each vector by its Euclidean norm $\sqrt{2}$:

$$u_1 = (1/\sqrt{2})[1, 1]^T \approx [0.7071, 0.7071]^T$$

$$u_2 = (1/\sqrt{2})[1, -1]^T \approx [0.7071, -0.7071]^T$$

NumPy np.linalg.eig(A) gives matching unit eigenvectors up to a sign (±), which is expected: both v and −v describe the same eigenvector line. The equation Av = λv was verified numerically for each pair.

AI Insight | An eigenvalue is defined by the existence of at least one nonzero vector satisfying (A − λI)v = 0. It can have infinitely many eigenvectors, including every nonzero multiple of any one eigenvector.

Mission 4 — Geometric Interpretation

Eigenvalue	Effect on length	Direction
3	Triple the length	Preserved
0.5	Half the length	Preserved
−2	Double the length	Reversed

For λ = 3 the vector stretches threefold, for λ = 0.5 it contracts to half, and for λ = −2 it stretches twofold while pointing in the opposite direction.

AI Insight | The magnitude |λ| controls scale; its sign controls orientation. λ = 0 sends its eigenvector to the zero vector, which has no direction.

Mission 5 — Matrix Diagonalization

Construct P from the manually obtained eigenvectors in columns, and put the matching eigenvalues on the diagonal of D:

$$P = [\,1 \quad 1\,;\,1 \quad -1\,] \qquad D = [\,3 \quad 0\,;\,0 \quad 1\,]$$

Since det(P) = −2 ≠ 0, the inverse exists:

$$P^{-1} = [\,0.5 \quad 0.5\,;\,0.5 \quad -0.5\,]$$

$$PD = [\,3 \quad 1\,;\,3 \quad -1\,]$$

$$PDP^{-1} = [\,2 \quad 1\,;\,1 \quad 2\,] = A$$

NumPy confirms np.allclose(A, P @ D @ np.linalg.inv(P)) is True. Diagonal matrices are easier to work with because powers and many functions can be computed element by element on the diagonal. For example, $A^k = PD^kP^{-1}$.

AI Insight | Diagonalization reveals independent directions along which a transformation acts by simple scaling. It supports matrix analysis and is closely related to PCA, where eigenvectors of the covariance matrix give principal directions.

Mission 6 — Eigenvalues in AI: PCA

1. Why PCA? Principal Component Analysis reduces feature dimensionality while preserving as much data variance as possible. It can simplify computation and visualization.

2. Why eigenvalues? Each eigenvalue of the covariance matrix measures variance along the corresponding principal component.

3. Why sort eigenvalues? Sorting from largest to smallest ranks the directions by how much variance they explain.

4. Which eigenvectors? Select eigenvectors associated with the largest eigenvalues, according to the chosen number of components or cumulative explained-variance threshold.

5. What if the largest is removed? The dataset may lose its dominant source of variation and reconstruction quality may deteriorate. High variance is not necessarily the same as predictive relevance.

Worked PCA Illustration

The notebook generates a reproducible synthetic dataset of 200 observations with two correlated features (random seed 42). After centering, it calculates the covariance matrix and uses np.linalg.eigh to obtain its eigenpairs, sorted in descending order.

$$\text{Eigenvalues} \approx [5.1295,\ 0.2258]$$

$$\text{Explained variance, first component} \approx 5.1295 \,/\, (5.1295 + 0.2258) = 95.78\%$$

Retaining only the first component therefore preserves approximately 95.78% of total variance in this illustrative dataset; the result is specific to the generated sample, not a universal PCA threshold.

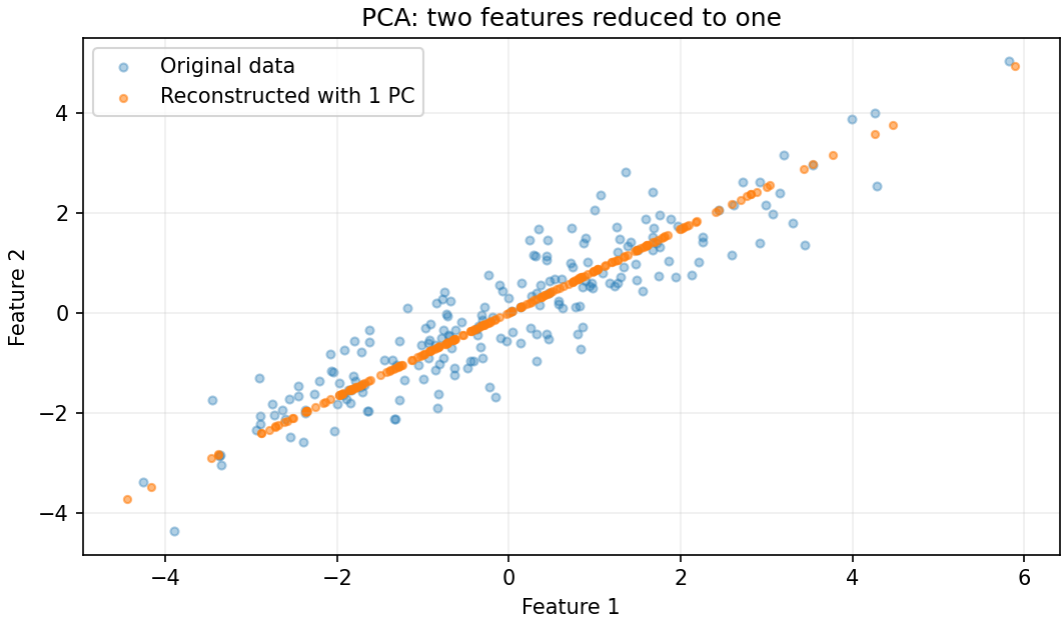


Figure 2. One principal component retains most variation in the synthetic data.

AI Insight | PCA uses covariance-matrix eigenvalues to quantify variance in different directions. Keeping eigenvectors with the largest eigenvalues allows reduction to fewer dimensions while minimizing squared reconstruction error for a given number of linear components.

Conclusion

The assignment demonstrates that eigenvectors identify transformation directions that remain on the same line, while eigenvalues quantify scaling and orientation. For A = [[2,1],[1,2]], the manual and NumPy results agree: eigenvalues 3 and 1; eigenvectors $[1,1]^T$ and $[1,-1]^T$. The matrix reconstructs exactly as PDP^{-1}. PCA extends the same ideas to identifying directions with maximum data variance for dimensionality reduction.